

CURIONEST

Acids & Bases Practice Pack

Theories, pH & pOH, and Acid Strength · 20+ Questions with Full Answer Key

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

What This Pack Covers

1. Identifying Brønsted–Lowry acids, bases, and conjugate pairs
2. Calculating pH, pOH, $[H_3O^+]$, and $[OH^-]$ at 25 °C
3. Using K_w and the relationship $K_a \times K_b = K_w$
4. Comparing acid strength using K_a and percent ionization
5. Recognizing common acid-base misconceptions

Reading Passage — Acids and Bases at a Glance

An acid is a proton (H^+) donor and a base is a proton acceptor, according to the Brønsted–Lowry definition. When an acid loses a proton, the species that remains is its conjugate base; when a base gains a proton, the product is its conjugate acid.

Water autoionizes: $H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$, with $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C. This single constant lets us convert between hydronium and hydroxide concentrations in any aqueous solution.

$pH = -\log[H_3O^+]$ and $pOH = -\log[OH^-]$; at 25 °C, $pH + pOH = 14.00$. Solutions with $pH < 7$ are acidic, $pH > 7$ are basic, and $pH = 7$ is neutral.

Strong acids (HCl , HNO_3 , H_2SO_4) ionize completely; weak acids (HF , CH_3COOH , HCN) ionize only partially. The ionization constant K_a measures strength — larger K_a means a stronger acid. The stronger the acid, the weaker its conjugate base.

★ **Common mistake:** Strength and concentration are different ideas: a dilute strong acid can have a higher pH than a concentrated weak acid.

★ **Common mistake:** $pH = 7$ is neutral only at 25 °C.

Quick Check — Multiple Choice (Questions 1–10)

1. Which species is the conjugate base of H_2O ?
 (A) H_3O^+
 (B) OH^-
 (C) H_2
 (D) O_2^-
2. Which of the following is a Brønsted–Lowry acid in the reaction $HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq) + OH^-(aq)$?
 (A) HCO_3^-
 (B) H_2O
 (C) H_2CO_3
 (D) OH^-
3. Which species is amphoteric?
 (A) OH^-
 (B) NH_4^+
 (C) HSO_4^-
 (D) Cl^-
4. At 25 °C, a solution has $pH = 3.00$. What is $[H_3O^+]$?
 (A) 3.0 M
 (B) 1.0×10^{-3} M
 (C) 1.0×10^{-11} M
 (D) 3.0×10^{-7} M
5. A solution has $[OH^-] = 1.0 \times 10^{-9}$ M at 25 °C. What is its pH?
 (A) 5.00
 (B) 9.00
 (C) 7.00
 (D) 14.00

6. Which acid is the strongest?
- (A) HCN ($K_a = 6.2 \times 10^{-10}$)
 - (B) CH_3COOH ($K_a = 1.8 \times 10^{-5}$)
 - (C) HF ($K_a = 6.8 \times 10^{-4}$)
 - (D) HNO_2 ($K_a = 4.0 \times 10^{-4}$)
7. K_b for the conjugate base of a weak acid with $K_a = 1.0 \times 10^{-6}$ is:
- (A) 1.0×10^{-6}
 - (B) 1.0×10^{-8}
 - (C) 1.0×10^{-14}
 - (D) 1.0×10^{-20}
8. Which statement about strong acids is TRUE?
- (A) They always have $\text{pH} < 0$
 - (B) They ionize completely in water
 - (C) They are always concentrated
 - (D) They have very small K_a values
9. Pure water at 25°C has:
- (A) $[\text{H}_3\text{O}^+] > [\text{OH}^-]$
 - (B) $[\text{H}_3\text{O}^+] < [\text{OH}^-]$
 - (C) $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7} \text{ M}$
 - (D) $\text{pH} = 0$
10. The pH of a solution changes from 4.00 to 5.00. The $[\text{H}_3\text{O}^+]$ has:
- (A) Doubled
 - (B) Increased $10\times$
 - (C) Decreased $10\times$
 - (D) Stayed the same

Calculate — Show Your Work (Questions 11–16)

11. Calculate $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ in pure water at 25°C .
12. A solution has $[\text{H}_3\text{O}^+] = 2.0 \times 10^{-6} \text{ M}$. Find $[\text{OH}^-]$ and classify the solution.
13. Calculate the pH of 0.050 M HCl (ionizes completely).
14. Calculate the pOH and pH of 0.010 M NaOH .
15. What are $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ when $\text{pH} = 8.50$?
16. A 0.10 M solution of a weak acid is 1.5% ionized. Calculate K_a .

Short Answer & Think (Questions 17–20)

17. Write the reaction of NH_3 with water and label the acid, base, conjugate acid, and conjugate base.
18. Explain why the conjugate base of a weak acid is a stronger base than the conjugate base of a strong acid.
19. A student says 'pH 2 is twice as acidic as pH 1.' Explain why this is wrong and give the correct relationship.
20. Why is water called amphiprotic? Give one reaction showing water as an acid and one showing water as a base.

Key Terms

Term	Definition
acid	Proton (H^+) donor in the Brønsted–Lowry definition
base	Proton (H^+) acceptor in the Brønsted–Lowry definition
conjugate pair	Two species that differ by exactly one proton
amphiprotic	Can act as both proton donor and proton acceptor
K_w	Ion-product constant of water = 1.0×10^{-14} at 25°C
pH	$-\log[\text{H}_3\text{O}^+]$
K_a	Acid ionization constant; larger value = stronger acid

Answer Key

All answers with explanations are embedded in the questions above — use the practice sections for the complete key.

Quick Check — Multiple Choice (Questions 1–10)

1. OH[−] (water minus one proton)

Why: Removing a proton (H⁺) from H₂O leaves OH[−].

2. H₂O (it donates a proton to HCO₃[−])

Why: H₂O loses H⁺ to become OH[−], so water is the acid.

3. HSO₄[−] (can donate OR accept a proton)

Why: OH[−] can only accept a proton; NH₄⁺ can only donate; Cl[−] does neither.

4. 1.0×10^{-3} M (pH = $-\log[\text{H}_3\text{O}^+]$)

Why: $10^{-3.00} = 1.0 \times 10^{-3}$ M.

5. 5.00 (pOH = 9.00, pH = 14 − 9 = 5)

Why: $[\text{H}_3\text{O}^+] = K_w/[\text{OH}^-] = 1.0 \times 10^{-5}$ M → pH = 5.00.

6. HF (largest K_a)

Why: Larger K_a means more ionization, hence a stronger acid.

7. 1.0×10^{-8} (K_b = K_w/K_a)

Why: $K_b = 1.0 \times 10^{-14} / 1.0 \times 10^{-6} = 1.0 \times 10^{-8}$.

8. They ionize completely in water

Why: Complete ionization is the definition of a strong acid; concentration is a separate idea.

9. $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7}$ M

Why: Autoionization produces equal amounts of both ions.

10. Decreased 10×

Why: Each pH unit = a tenfold change; higher pH = lower $[\text{H}_3\text{O}^+]$.

Calculate — Show Your Work (Questions 11–16)

11. $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7}$ M ($x^2 = K_w$)

12. $[\text{OH}^-] = 1.0 \times 10^{-14} / 2.0 \times 10^{-6} = 5.0 \times 10^{-9}$ M. Acidic.

13. pH = $-\log(0.050) = 1.30$

14. pOH = 2.00; pH = 12.00

15. $[\text{H}_3\text{O}^+] = 10^{-8.50} = 3.2 \times 10^{-9}$ M; $[\text{OH}^-] = 1.0 \times 10^{-14} / 3.2 \times 10^{-9} = 3.1 \times 10^{-6}$ M

16. $[\text{H}_3\text{O}^+] = 1.5 \times 10^{-3}$ M; $K_a \approx (1.5 \times 10^{-3})^2 / 0.10 = 2.3 \times 10^{-5}$

Short Answer & Think (Questions 17–20)

17. NH₃(aq) + H₂O(l) ⇌ NH₄⁺(aq) + OH[−](aq). Acid: H₂O · Base: NH₃ · Conjugate acid: NH₄⁺ · Conjugate base: OH[−]

18. The weaker the acid, the less it ionizes, so its conjugate base has a greater tendency to accept a proton back ($K_a \times K_b = K_w$; smaller K_a → larger K_b).

19. pH is a log scale: pH 1 has $[\text{H}_3\text{O}^+] = 10\times$ that of pH 2, so pH 1 is TEN times more acidic, not twice.

20. Water can donate a proton (e.g., H₂O + NH₃ → NH₄⁺ + OH[−]) and accept a proton (e.g., HCN + H₂O → CN[−] + H₃O⁺).